Rethinking consent in human-machine interactions: beyond yes and no models

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Abstract

This paper addresses the growing crisis of consent in machine interactions, particularly in data-driven technologies where consent is often legally compliant but ethically shallow. Users frequently accept default settings and terms without understanding, undermining the notion of informed, voluntary consent. Drawing from theories of relational consent—originating in the field of sexuality—we explore more dynamic, ongoing, and context-aware models of consent in computing. This challenges static, one-time consent frameworks and reframes consent as a dialogic, ethically engaged process. We present the notion of relational consent in computer interaction and apply it through exploration and speculative design in two case studies. The first examines animal husbandry, where power asymmetries and datafication challenge notions of agency. The second explores video games, reimagining consent as continually negotiable. Together, these cases offer highly diverse perspectives that illustrate a wide range of consent-related challenges. The findings aim to better support autonomy, integrity, and fairness in machine-mediated interactions. This work contributes to the design of more ethical technologies that take into account both human and more-than-human considerations.

Keywords

Digital consent, speculation, more-than-human, design, critical HCI

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1 Introduction

People constantly consent to human-machine interactions when using digital technology and services. This is most acute in contemporary algorithm-driven technology, which relies on personal data to function. Much of this consent-giving is legally regulated and legally compliant. Regulatory laws often aim to protect users from having their data misused and/or their integrity violated. Yet, there is a vast abundance of research showing how people when given choices like this will pick the default alternative, accept cookies without any understanding of what they mean, and accept any agreement of use without reading it (1% read Terms of Service agreements says studies) [10, 16, 18, 23]. These choices might be legally binding, but do not fit how we generally understand consent: as a voluntary, informed act [3]. Without information and, importantly, understanding, we cannot properly consent [14]. In short, research univocally states that whatever it is we do, it is not consenting.

This paper, which draws on a new research project, addresses this crisis of consent in computing by speculative engagement, building on relational consent principles to push our current understanding of what machine consent is and what it could be. We investigate consent through different cases that shed light on the ethical and practical implications of relational consent, both equal and unequal. More specifically, the paper aims at providing a starting point for exploring how critical and pathbreaking theories of *relational consent* can provide new ways of understanding human-machine consent, further informing the development of novel practices that support ethical human-centric interactions and data practices which safeguard people's integrity, autonomy and data in a time of crisis. One important point of departure is theories of relational consent in the field of sexuality, which others have argued open new ways forward for designing consent in and with technologies [15]. These theories of relational consent argue that we should not see consent as a one-time, static act, but an ongoing, dynamic process that requires active participation, respect, and understanding from all parties involved [2, 4, 12, 25]. We argue that this nuanced understanding of consent as a social act also has important implications for consent in human-machine interactions. This is key because how we understand consent directly influences the solutions we propose to prevent harm [25].

After setting out a framework for relational consent in computer interactions, this paper presents preliminary results from two critical studies of technological, data-driven contexts where we examine how relational consent theories can address the crisis of consent in machine interactions. Methodologically, we engage in "what if" speculation in that we explore through physical and digital design or storytelling and questioning to reimagine alternative social, political, and technological realities [6]. First, we explore animal hubandry as an asymmetric situation where the datafied can neither agree nor disagree and speculate on what relational consent could mean in this context. Secondly, we detail two games speculating on the opportunity for relational consent to various types of violence in video games in the form of constantly being able to change consent, but still be able to engage in play. These cases are tied together by the central question: *What alternative consent structures can we imagine with relational consent to support fairer interaction with machines and each other through machines?*

2 Developing a framework for relational consent in human-computer interaction

The increasing value of data has driven consent practices to become opaque and more complex. In this context, research on consent and data in machine interactions has focused on how users give, manage, and withdraw consent for the collection, use, and sharing of personal data by machines, especially AI and algorithm-driven digital platforms [16, 18]. Although early understandings of consent emphasised explicit, informed consent through clear communication, scholars have highlighted a privacy paradox; people value privacy but make decisions that go against this due to limits in decision-making, lack of understanding, bad design, and the normalisation of data sharing (for a review see [9]). Current digital consent structures further help cement a business model where users trade personal data for access to information or services, and this model has come to be seen as the only alternative [5]. Recent advances in AI and the way that AI is trained on our data have sharpened this issue.

To deepen the consent design space, we have turned to the field of sexuality. This area is at the leading edge of thinking about consent, and can arguably help move beyond current limitations in work on digital consent. Previous work on consent in the sexual and relational domain, we argue, has come in three waves. First, the "No means No model" sought to address unwanted sexual advances. Here, consent was implied unless someone clearly stated no [11]. After critique, foremost from feminist and anti-rape organisations, a new affirmative model of consent arose. Friedman and Valenti [8] popularised the notion of affirmative consent, or the "Yes means Yes" model. Here, consent is not assumed and is seen as an act in itself. The lack of yes means no. The 2016 FRIES model from Planned Parenthood in the US bolstered this affirmative consent model by defining consent as Reversible, Informed, Enthusiastic, and Specific. Previously, scholars have suggested that the FRIES model provides a provocative insight into how we can rethink machine consent [22].

Yet, the focus on consent as something to be communicated has, in turn, received critique as the focus on teaching people how to communicate place a great burden on potential victims to be able to and daring to withdraw consent once given and having enough communication skills to be able to enact affirmative consent [26]. We argue that we have entered a third wave where new models of consent are drawn from queer and relational models. Relational consent highlights the importance of ongoing negotiation, respect for boundaries, and clear communication in all interactions, whether between individuals or with systems that impact our lives. Relational consent builds on the yes means yes model, yet allows us to highlight the complexity of consent, which can move beyond no or yes acts. Seeing consent as relational [12] reframes it to be more nuanced, dialogic, and flexible rather than stable and binary, i.e., simple no or yes statements.

3 Reimagining computing after the consent crisis

In the remainder of this paper, we reflect on our ongoing work that explores future possibilities for relational consent in computing. We aim to provoke debate, raise questions, and challenge preconceptions about potential technological futures [6]. Through speculation, we are questioning societal expectations of technological development and influencing that development by highlighting both desired and undesired futures [1]. We start with the question "what if" and include a detailed analysis of the societal, ethical, and practical consequences of the speculation. For the two cases detailed here, we are questioning and through this reimagines current consent practices.

3.1 Animals and consent

We have conducted a pilot study examining the notion of consent in relationships characterised by profound power asymmetries, most notably, those between humans and animals. This imbalance raises critical ethical questions around agency and autonomy in technology use. Nowhere is this more evident than in domains such as animal husbandry, where humans often design technological interventions without consideration for the affected non-human participants' ability to comprehend or influence outcomes.

Beyond practical concerns, we want to use this case to highlight broader ethical questions around consent and extreme datafication. It provides a lens for us to reconsider what decision-making includes when individuals—here animals—cannot understand the implications of their actions or the systems they are embedded within [13]. The field of animal-computer interaction explores this complexity, developing technologies for both companion animals and livestock. However, technologies for pets are frequently designed to foster emotional bonds and mutual understanding, while those in animal husbandry are primarily focused on efficiency and productivity [17]. Given these differences, our ongoing study concentrates specifically on the husbandry context, where ethical tensions are arguably more pronounced. The study aligns with the field of more-than-human-centred design, which draws attention to what it means to centre non-human users in design [19, 24]. Through the lens of more-than-human-centred design, we re-examine the concept of consent, not as a binary state, but as a nuanced ethical relationship. Our inquiry attempts to understand how agency unfolds or is denied in environments where decisions are shaped by systems beyond the user's grasp.

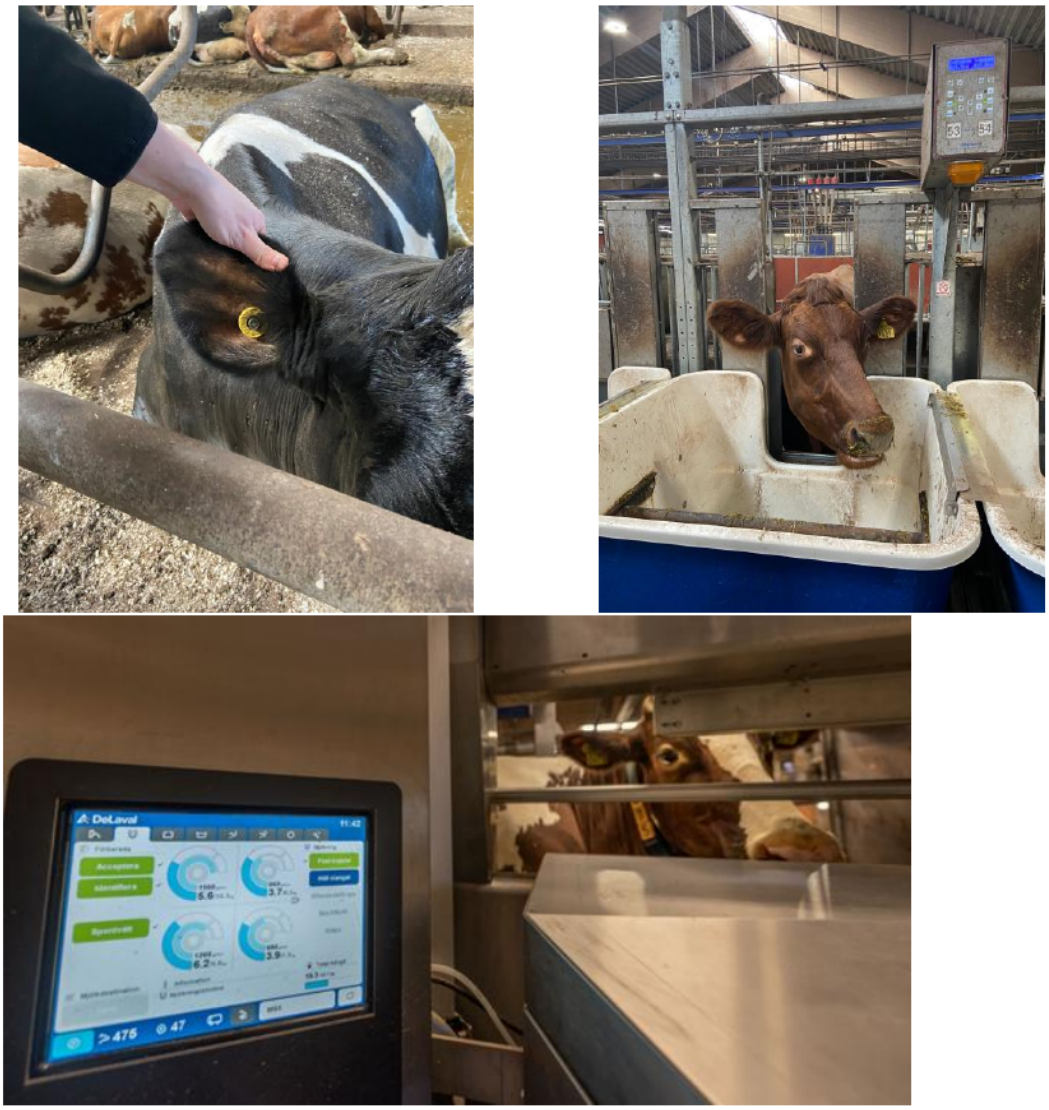


Figure 1: Image 1, 2 and 3: Three pictures from a modern dairy farm that attach sensors in the ears that in turn datafies movement, how much each cow eats, and their individual milk production on each of the four teats, the upmost pictures taken by us, the last one taken from Lantbruksnytt from the same farm.

In the fieldnotes in this ongoing study, we wrote:

> "This is not a world I am familiar with. It's easy to remain blind to the production side of our food systems. But walking through this facility, what strikes me most is how thoroughly the cows are connected to technology. Their world is almost entirely datafied, and metrics are generated for nearly every aspect of their existence: health, productivity, behaviour. The only thing left unmeasured, it seems, is their poop... While the cows are still firmly under the control of their human owners, there is a paradox at play. The milk robots seemingly give the cows more freedom, allowing them to choose when to be milked, for example. Yet, through this same system, they seem even more tightly bound, more deeply integrated into a regime of constant measurement and control."

In our study on milking robots, cows that repeatedly resist or fail to interact successfully with the robots are often removed from the production and sent to slaughter, a rather brutal consequence that underscores the utilitarian logic behind these technologies. This raises further ethical questions: Does the inability to conform to a technological system equate to a lack of worth? How do we think about the needs of human-centred systems with the lived experience of the non-humans they depend on?

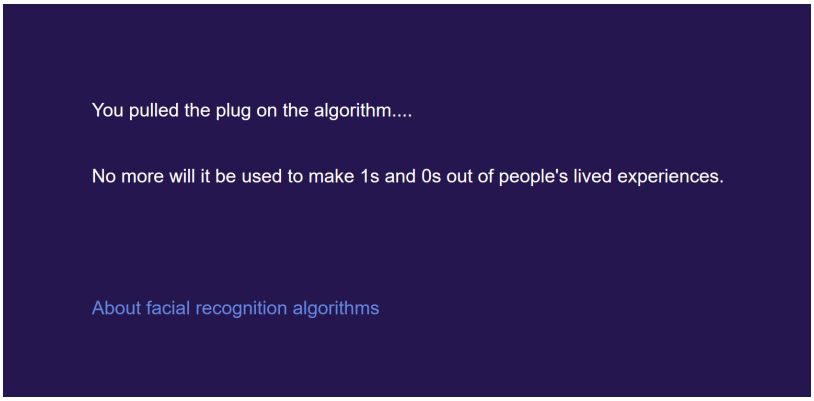


Figure 2: Screenshot from *The GenderCrusher* showing the destroy end.

Speculating on consent for those who can neither consent nor refuse raises nuanced ethical questions; urging a rethinking of technological consent, not as a one-size-fits-all standard, but as a relational, contextual, and ethical process, one that accounts for agency, resistance, and the limits of comprehension.

3.2 Relational consent in video games

Our other case concerns a completely different environment: video games. A well-known consent problem in games is that players are rarely able to resist in-game actions or choices that go against their desires without quitting the game. As designed experiences, games are paradoxically choice machines with no true choices [7]. If a player objects to shooting an enemy, missing on purpose and running away is often not possible. To play a digital game, a player must surrender to the game [20]; they have to consent to the rules as they are encoded. Resistance can only take place through refusal to engage, and saying "Yes" in turn means playing the game as is. Below, we detail two working designs where we explore what relational consent in a game experience might imply. Where the player's surrender is part of a relational consent negotiation

In the *GenderCrusher*, users play at training a facial recognition AI to be more inclusive of Queer identities. Through the design, players can get a sense of the vast range of human identities and the difficulty in external actors defining who someone is. The game deals with the sensitive issue of identity and labelling of others, perhaps against their desires. To ensure that players always have the opportunity to renegotiate their consent, without quitting the game, a big red panic button was installed. Pressing the button is possible at any time, and doing so means players come to an end where the algorithm is destroyed. An end that reflects their choice to not say no to the game by walking away, but neither saying yes to the activities of the game. While the experience is built upon people completing the classification, the red button allows playing and taking part in the experience, without consenting to the violence of gender classification. Pressing the button means succeeding in the game, yet in a way that reflects the choice of forcing a stop to the algorithm.

In *P-our love is the light that fuels the light*, players interact with a future imagination of a natural language chatbot that is presented as



Figure 3: Photo from the hybrid experience P in the process of being plugged in.

built by the player out of partly digital, partly biological components. To start, players have to plug in the machine.

There is always an option not to plug in *P*. If so, players are taken to an ending. They finish the game without having to start the AI. From this ending, they can watch the credits, or change their mind and plug it in. Choices are reversible. If a player chooses to play but then desires not to allow the AI they have trained to continue to exist, they can revoke consent. Here as well, the game responds to the revoking of consent, to players pulling the plug, and offers an alternative experience. In the case of *P*, players can also change their mind and plug in the experience again, and so continue from that point. Saying yes or no is dynamic and changeable, and saying no does not mean being locked out of an experience.

4 What if? Initiating speculation about consent

Through these speculations, we have initiated the next step. Based on the dimensions of consent that we have explored, we imagine that consent could include aspects such as being ongoing, partial and dynamic, requiring some kind of active participation, respect, and understanding. In our cases, participants can change their minds and consent to some parts of the experience, but not others.

Both no and yes can exist in their choices. This opens up questions about what new ways of interacting to complement those we already have.

Among animals, consent has traditionally been assessed through behavioural cues, discomfort, distress, or avoidance are taken as signals that an animal does not "consent" to an activity [21]. However, in an increasingly data-driven world, this approach may no longer suffice. Is immediate comfort an adequate ethical metric, or must we also consider the long-term implications of datafication, bodily surveillance, and performance optimisation? How can we extend ethical consideration to include the best interests of non-consenting users, both animals and humans, particularly when those interests are profoundly entangled in algorithmic systems?

For games, a relational perspective allows us to think about what nuanced and constantly ongoing consent mechanisms [15] we can imagine. In digital services, we often have to consent to be users. Are there ways to break this design structure? What if we refuse and still want to be users?

Finally, relational consent has offered us critical new ways to ask questions about and shed light on what we call an ongoing consent crisis. There is a fundamental gap in how we approach consent in the digital age. In the context of machine interactions, responsibility should shift to ensuring machines respect consent. Violations of consent, whether they occur socially, politically, or legally, need to be fully understood to mitigate harm effectively. Our goal forward is to push the boundaries of what it means to support consensual human-machine interactions. To provide a new critical reading of consensual machine interactions that push beyond current models.

5 Concluding remarks

For us, exploring critical insights from the field of sexual consent has been particularly productive for emphasising the challenges users face in navigating current technological consent situations, especially when interacting with AI or automated systems. Applying relational consent in our project has supported a more nuanced, dynamic, and ongoing understanding that considers power dynamics, user autonomy, and the evolving nature of consent in digital and technological interactions. An understanding of consent beyond yes or no.

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